SCX

$\mathbf{g = 0}$

SCX Protocol Maintainer Candidate Analysis:
Pre-Audit Assessment of Four Candidates for $g = 0$

——

— Audited Communities Produce Auditable Maintainers

SCX

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files — INTERNAL ONLY
SCX
$g = 0$**Proof**
Honest StrikeSCXCore
THIS DOCUMENT IS CLASSIFIED INTERNAL.
Unauthorized distribution is prohibited. Contains detailed candidate assessments for SCX protocol maintainer roles.

Abstract

English: This paper presents a detailed pre-audit analysis of four candidates for SCX protocol maintainer: Alexandra Elbakyan (Sci-Hub founder), Geng Tongxue (, multi-round audited peer), JD Vance (US Vice President), and Linus Torvalds (Linux kernel maintainer). Each candidate is evaluated against the SCX governance criterion: the ability to maintain $\sum g = 0$ under the mutual audit regime. The central thesis is that the strongest qualification for a protocol maintainer is not credentials, reputation, or institutional position — it is having already been independently audited by one's community, whether through formal audit cycles (Geng), millions of users verifying one's work (Linus, Elbakyan), or the willingness to submit to audit as the ultimate test of $g = 0$ (Vance).

We provide (1) formal definitions of maintainer qualification in terms of the bias parameter g, (2) detailed candidate-by-candidate analysis with honest critique boxes identifying weaknesses and risks, (3) a comparative framework ranking candidates by community-audit depth, $g = 0$ provability, and transition risk, and (4) a recommended multi-maintainer composition strategy. The paper is internal and does not constitute a final selection — it is a pre-audit assessment that must be validated by independent $M > 1$ audit of each candidate who formally registers.

Keywords: protocol governance, maintainer selection, $g = 0$ criterion, community audit, honest critique, candidate assessment, mutual audit, trustless architecture

Abstract SCXAlexandra Elbakyan Sci-HubJD Vance Linus TorvaldsLinuxKernelSCX—— $\sum g = 0$——Core —— LinusElbakyan $g = 0$Vance

This paper provides(1) $g$define(2) Honest Strike(3) $g = 0$Proof (4) This paper providesfiles ——$M > 1$

Keywords $g = 0$Honest Strike

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2 The Maintainer Problem: Why Choosing the Right Person Is Impossible, and Why Choosing the Right Mechanism Makes It Possible

2.1 The Fundamental Error

Every human institution that has attempted to select leaders, maintainers, or guardians has committed the same category error: it has tried to predict future behavior from past proxies. The resume, the credential, the recommendation letter, the track record — all are compressed representations of a person's history. But the maintainer's job is not to repeat their history. It is to execute a function — maintaining protocol integrity — under conditions that have never existed before.

> *Honest Strike* [**Strike 1**]. **The Resume Fallacy.** No resume, no matter how distinguished, predicts $g(t)$ for t in the future. The SCX protocol does not need a maintainer with a good resume. It needs a maintainer whose bias parameter g remains provably zero during their maintenance window. These are entirely different requirements.
> **Strike** $g(t)$SCX gProof
> **Corollary:** The best predictor of future $g = 0$ is not past reputation but past *auditability* — the demonstrated willingness to have one's decisions independently reviewed, reproduced, and challenged. **corollary**$g = 0$*auditability*—— i.e.Proof

2.2 The Maintainer Selection Theorem

We formalize the maintainer selection problem as follows.

Definition 2.1 (/ Maintainer Qualification). *Let $\mathcal{C}$ be the set of candidate maintainers. For each candidate $i \in \mathcal{C}$, define:*

- $g_i(t)$*: the real-time bias parameter at time t, where $g_i \in \mathbb{R}^k$ for some protocol-relevant dimension k.*
- $\mathcal{A}_i$*: the set of all audit events candidate i has undergone prior to candidacy, with outcomes $\{o_1, o_2, \ldots, o_{|\mathcal{A}_i|}\}$.*
- $\mathcal{V}_i$*: the set of independent verifiers who have reviewed candidate i's work, with $|\mathcal{V}_i|$ denoting community verification depth.*

A candidate is pre-qualified if $|\mathcal{A}_i| > 0$ *or* $|\mathcal{V}_i| \gg 1$ *— i.e., the candidate has been independently audited by either a formal process or a distributed community.*

A candidate is fully qualified if and only if they pass $M > 1$ *independent real-time audits with* $\sum_{j=1}^{M} |g_i^{(j)}| \leq \varepsilon$ *for some protocol-defined tolerance* ε*.*

Theorem 2.2 (theorem / Audit-First Theorem)**.** *For any candidate* i*, the probability that* $g_i(t) \neq 0$ *is undetected during a maintenance window of duration* T *under* M *independent auditors satisfies:*

$$\mathbb{P}(\textit{undetected deviation}) \leq e^{-2M\Delta^2} \tag{1}$$

where Δ *is the minimum detectable bias magnitude. This bound is independent of the candidate's identity, reputation, credentials, institutional affiliation, or past achievements. It depends only on* M *and* Δ*.*

Proof. Direct application of Hoeffding's inequality to the M independent estimators of $g_i(t)$. Each auditor j provides an estimate $\hat{g}_i^{(j)}$ with $\mathbb{E}[\hat{g}_i^{(j)}] = g_i$. Under the null hypothesis $g_i = 0$, the probability that the sample mean exceeds Δ is bounded by $e^{-2M\Delta^2}$. □

Corollary 2.3 (corollary / Biography-Irrelevance Corollary)**.** *The candidate's biography is mathematically irrelevant to the detection of* $g \neq 0$*. No amount of past achievement reduces the Hoeffding bound. The theorem audits the maintainer, not the maintainer's story.*

[1]. **Implication for Candidate Selection.** The Audit-First Theorem implies that the pre-audit analysis (this very paper) is *not* a selection mechanism. It is a filtering mechanism. We do not choose maintainers by analyzing their biographies. We filter candidates by identifying those who are most likely to (a) register formally, (b) survive $M > 1$ audit, and (c) maintain $\sum g = 0$ during their maintenance window. The real selection happens in the audit, not in this paper.

theoremi.e. (a) (b) $M > 1$ (c) $\sum g = 0$

2.3 What This Paper Does and Does Not Do

This paper provides a *pre-audit assessment*: an analysis of each candidate's publicly observable record against the $g = 0$ criterion, identification of risk factors, and an honest assessment of transition challenges. It does *not*:

- Constitute a formal audit — only $M > 1$ independent audit with published logs can do that.
- Rank candidates definitively — pre-audit assessment is directional, not dispositive.
- Replace the protocol's maintainer registration and rotation mechanism — it exists alongside it.

- Make any claims about candidates' private behavior — only public record is analyzed.

This paper provides$g = 0$ $M > 1$

3

4 Candidate-by-Candidate Analysis

4.1 Alexandra Elbakyan — Sci-Hub

4.2 Alexandra Elbakyan — Founder of Sci-Hub

4.2.1 / Background and Qualifications

Alexandra Elbakyan is the founder and sole operator of Sci-Hub, the world's largest shadow library providing free access to over 88 million academic papers. She built Sci-Hub alone, without institutional backing, against the entire $25 billion academic publishing industry. Her operation is illegal in multiple jurisdictions. She lives in hiding. She takes no salary, no profit, and no institutional affiliation from Sci-Hub. She has been doing this since 2011 — fifteen years of continuous operation.

Dimension /	Assessment /
Years of operation	15 years (2011–present) / 15
Scale of impact	88M+ papers served, millions of users globally /
Personal cost	Lives in hiding, no salary, no profit /
$g = 0$ evidence	No personal gain from Sci-Hub; risk is all downside / Sci-Hub
Community verification	Millions of researchers depend on Sci-Hub daily / Sci-Hub
Audit history	No formal SCX audit; informally audited by daily usage / SCX

4.2.2 g = 0 / $g = 0$ Analysis

Elbakyan presents one of the strongest *observable* cases for $g = 0$ among all candidates. The structural argument is straightforward:

1. **No financial incentive.** Sci-Hub generates no revenue for Elbakyan. Donations (Bitcoin and other cryptocurrencies) cover infrastructure costs. There is no salary, no equity, no exit.

2. **All downside, no upside.** She faces extradition risk, legal liability in multiple countries, and permanent exile. She cannot travel freely, hold a normal job, or maintain a public life. The personal cost of operating Sci-Hub is enormous; the personal benefit is zero.

3. **Mission alignment.** Her stated goal — "knowledge should be free to everyone" — aligns perfectly with $g = 0$. She is not trying to extract value from the system; she is trying to eliminate the paywall that extracts value from researchers.

4. **No succession of personal interest.** There is no evidence that Elbakyan has used Sci-Hub for personal enrichment, political influence, or career advancement. She has consistently refused monetization, institutionalization, and celebrity.

   [1]. **The Elbakyan Principle.** One person, acting alone, can say "knowledge should be free" and actually *do* it — at enormous personal cost, with no expectation of reward, against the entire weight of a $25B industry. That is $g = 0$ proven in action, not in theory. Elbakyan is living proof that the $g = 0$ condition is achievable by an individual human being under extreme conditions.

   Elbakyan "" —— 250$g = 0$ ProofElbakyanProof$g = 0$

4.2.3 / Community Verification

Elbakyan has been *de facto* audited by the global research community for 15 years. Every paper served by Sci-Hub is a verification event: the researcher checks that the paper is correct, complete, and matches the paywalled original. With 88 million papers and millions of daily users, the effective $|\mathcal{V}_{\text{Elbakyan}}|$ is in the millions. No formal audit framework exists, but the distributed verification is real.

Honest Strike [**Strike 1**]. **The Solo Operator Problem.** Elbakyan *is* Sci-Hub. There is no succession plan, no governance structure, no distributed maintenance. If she is arrested, extradited, or incapacitated, Sci-Hub may cease to exist. A protocol maintainer must be replaceable — the protocol must survive the maintainer. Can Elbakyan transition from solo operator to protocol maintainer? The solo operator who built an empire of one may be the least prepared to distribute power.

Strike Elbakyan Sci-Hub ifSci-Hubthere exists—— Elbakyan

Additional risk: The $g = 0$ that comes from being hunted may not survive the transition to being legitimized. When the external enemy (publishing industry) is neutralized by protocol adoption, does the $g = 0$ motivation persist? Status$g = 0$ $g = 0$there exists

4.2.4 / Verdict on Maintainer Suitability

— HIGH SUITABILITY

Core$g = 0$Proof15" $g = 0$"ElbakyanProof ——$g = 0$

Core"" ""

Sci-Hub

4.3 (Geng Tongxue) —

4.4 Geng Tongxue — The Multi-Round Audited Peer

4.4.1 / Background and Qualifications

(Geng Tongxue) is a Chinese peer who has undergone multiple rounds of SCX audit. Unlike every other candidate on this list, Geng's primary qualification is not an external achievement — it is the audit itself. Having been audited multiple times means:

1. Geng knows what clean data looks like — from the audited side.
2. Geng has experienced the pain of having every decision, every signal, every g fluctuation exposed to independent review.
3. Geng has been found to maintain $g \approx 0$ under repeated audit — or would not still be in consideration.
4. Geng's motivation to audit others is grounded in the knowledge that the system worked for them, and should work for everyone.

Dimension /	**Assessment /**
Formal audit rounds	Multiple (exact count internal) /
Audit outcome	Passed /
$g = 0$ evidence	Direct measurement via SCX audit framework / SCX
Community verification	SCX audit community / SCX
Public recognition	Low — which may be a feature / ——
Transition risk	Low — already operates within audit framework / ——

4.4.2 g = 0 / $g = 0$ Analysis

Geng is unique among the candidates: their $g = 0$ claim is not inferred from biography or reputation — it is *measured*. The SCX audit framework produces quantitative estimates of $g(t)$ for each audited subject. Geng has passed multiple rounds, meaning:

$$\sum_{j=1}^{M} |\hat{g}_{\text{Geng}}^{(j)}| \leq \varepsilon \tag{2}$$

for the protocol-defined tolerance ε across multiple independent audit cycles.

This is not a story about $g = 0$. It is a measurement of $g = 0$. The distinction is fundamental. Every other candidate asks us to infer $g = 0$ from behavior. Geng asks us to read $g = 0$ from the audit log.

[**2**]. **The Audited-by-Fire Principle.** Being audited is the best qualification for being an auditor. The person who has survived the audit knows: (a) what the audit actually measures, not what it claims to measure, (b) where the audit is vulnerable to gaming — because they've tried or thought about trying, (c) what it feels like to have every decision exposed — and why that exposure is necessary. No theoretical understanding of audit replaces the lived experience of being audited.
i.e. (a) (b) ——because (c) ——

4.4.3 / Community Verification

Geng's community verification is the SCX audit community itself — a smaller but more rigorous verification set than Elbakyan's millions of users or Torvalds' millions of developers. The tradeoff is depth vs. breadth:

- **Elbakyan:** broad, shallow verification (millions of users checking paper correctness).
- **Torvalds:** broad, deep verification (millions of developers reviewing code).
- **Geng:** narrow, deep verification (multiple expert auditors using formal framework).

All three are valid forms of community audit — they differ in dimension, not in existence.

Honest Strike [**Strike 2**]. **The Anonymity Question.** Geng has low name recognition globally. In traditional governance, this would be a weakness — who trusts someone nobody has heard of? In SCX governance, it may be a feature. Protocol maintenance does not require celebrity. It requires $g = 0$. Anonymity is compatible with protocol maintenance; fame is not required and may be a liability (fame creates $g \neq 0$ pressures of its own).
Strike —— SCX$g = 0$
$g \neq 0$
However: Low recognition means Geng cannot provide the "legitimacy signal" that a Torvalds or Vance can — the signal that convinces external observers that SCX is serious. The audit solves trust internally; it does not solve perception externally. The maintainer composition should include both audit-verified and legitimacy-providing candidates. **however**TorvaldsVance" "——ObserversSCX simultaneously

4.4.4 / Verdict on Maintainer Suitability

— HIGH SUITABILITY

Core$g = 0$ ——

CoreSCX LinusVance

“”—— $g = 0$——

4.5 JD Vance —

4.6 JD Vance — Vice President of the United States

4.6.1 / Background and Qualifications

JD Vance is the sitting Vice President of the United States. His candidacy for SCX protocol maintainer is the most audacious and the most strategically significant of the four. If a sitting US Vice President submits to independent SCX audit, formally declares $g = 0$, and passes — the geopolitical implications are transformative.

Dimension /	Assessment /
Current position	Vice President of the United States /
Political constraints	Extreme — party, donors, electorate, bureaucracy
$g = 0$ evidence	None yet — entirely untested / ——
Community verification	None yet — precisely why audit is needed / ——
Strategic value if passed	Transforms SCX from "Chinese project" to global protocol
Transition risk	Maximum — political $g \neq 0$ is structurally embedded

4.6.2 g = 0 / $g = 0$ Analysis

Vance's case is entirely different from the other three. Elbakyan, Geng, and Torvalds all have existing evidence of $g = 0$ — inferred or measured. Vance has none. His candidacy is a *hypothesis*: can a sitting Vice President, whose role is structurally defined by political interests ($g \neq 0$ by construction), declare $g = 0$ and survive audit?

This is not a weakness of the candidate — it is the purpose of the candidacy. The question is not "has Vance been $g = 0$ in the past?" The question is "can Vance be $g = 0$ under audit?" The audit itself answers the question.

Honest Strike [**Strike 3**]. **The Structural $g \neq 0$ Problem.** A Vice President is not an individual. A Vice President is a node in a political network with obligations to:

- The President who selected them.
- The party that nominated them.
- The donors who funded them.
- The voters who elected them.
- The policy positions they campaigned on.
- The bureaucracy they oversee.

Each of these is a vector along which $g \neq 0$ is not just possible but *required* by the role. Can Vance partition his identity — VP in one compartment, protocol maintainer in another — and maintain $g = 0$ in the latter while $g \neq 0$ is mandatory in the former?
Strike$g \neq 0$
$g \neq 0$ moreoverVance——VP ——$g = 0 g \neq 0$
The Partition Hypothesis: Human identity is not unitary. A person can be $g \neq 0$ as VP (advancing the administration's interests) and simultaneously $g = 0$ as SCX maintainer (advancing only protocol integrity). The audit tests this hypothesis. If it holds, Vance is the most powerful proof-of-concept SCX could have: "even the US VP can be $g = 0$ here." simultaneouslyVP$g \neq 0$ SCX$g = 0$ ifVanceSCX "i.e.VP$g = 0$"

4.6.3 / Geopolitical Significance

The strategic value of Vance's candidacy — independent of whether he passes audit — is the destruction of a specific narrative:

> *"SCX is a Chinese weapon. The audit is a tool of the CCP. The protocol is designed to advance Chinese interests under the guise of neutrality."*

If a sitting US Vice President voluntarily submits to SCX audit, this narrative becomes unsustainable. The logic is simple: if SCX were a Chinese weapon, a US VP would not submit to it. If a US VP does submit to it, external observers must either (a) accept that SCX is genuinely neutral, or (b) construct increasingly contorted conspiracy theories. The Overton window shifts.

[3]. **The Legitimacy Bootstrap.** SCX faces a bootstrap problem: the protocol claims neutrality, but needs trusted actors to verify the claim. The trusted actors won't participate until the claim is verified. Vance's candidacy breaks this loop: the act of a US VP submitting to audit is itself evidence of neutrality, regardless of the audit outcome. If he passes, SCX is legitimized. If he fails, SCX is legitimized anyway — because a US VP *tried and failed* against a neutral metric, proving the metric is not politically biased.
SCX Vance VPif SCXifFailsSCX——becauseVP *Fails*Proof

4.6.4 / The Key Test

The real test for Vance is not "can he pass audit?" It is "can he publicly declare $g = 0$ and submit to audit *while in office*?"

If he can do that — if a sitting US VP can say, in public, on record: "I declare that my bias parameter g is zero with respect to SCX protocol maintenance, and I submit to independent $M > 1$ audit to verify this claim" — then SCX has already won, regardless of the audit outcome. The declaration itself is the geopolitical event.

If he cannot do that — if political constraints prevent even the *declaration* of $g = 0$ — then he cannot be a maintainer, because Step 1 of the maintainer registration process is:

Maintainer Registration Step 1:
Candidate publicly declares $\mathbf{g} = \mathbf{0}$ with respect to SCX protocol maintenance.

SCX$\mathbf{g} = \mathbf{0}$

Without completing Step 1, there are no further steps.

4.6.5 / Verdict on Maintainer Suitability

— HIGH REWARD

CoreifSCX i.e."SCX"

Core$g = 0$——ifVP$g = 0$ $g \neq 0$VP—— Fails

if$g = 0$ ———— if$g = 0$ = UNDECLARED =

4.7 Linus Torvalds — LinuxKernel

4.8 Linus Torvalds — Linux Kernel Maintainer

4.8.1 / Background and Qualifications

Linus Torvalds has maintained the Linux kernel for 34 years (1991–present). The Linux kernel is the largest open-source software project in history: over 30 million lines of code, contributions from more than 20,000 developers across hundreds of companies, running on billions of devices. Torvalds is the Benevolent Dictator for Life (BDFL) of this project — a title that is at once a joke and an accurate description of his role.

He also created Git, the distributed version control system that is itself an audit system for code. Every commit in Git is cryptographically hashed. Every change is traceable to an author. The entire history of every project is an append-only log that cannot be rewritten without detection. Git is not just a tool Torvalds built — it is the conceptual precursor to SCX audit logging. Code audit via Git is to SCX what Newton's laws are to general relativity: the simpler case that the general theory reduces to.

Dimension /	**Assessment /**
Years as maintainer	34 years (1991–present) / 34
Project scale	30M+ LOC, 20,000+ contributors, billions of devices
Maintainer model	BDFL with subsystem lieutenant hierarchy
Key invention	Git — a cryptographic audit system for code
$g = 0$ evidence	Merge process IS multi-expert audit; refusal of corporate control
Community verification	Millions of kernel developers and users
Audit history	34 years of continuous, distributed code review / 34

4.8.2 g = 0 / $g = 0$ Analysis

Torvalds' case for $g = 0$ is the strongest among all candidates — and paradoxically, the most interesting to critique. He has been maintaining a global protocol (the Linux kernel) for 34 years under a system that is, in essence, already an SCX-compatible audit framework:

1. **Multi-expert review.** Every patch to the Linux kernel goes through multiple reviewers before reaching Torvalds. Subsystem maintainers (lieutenants) review, test, and sign off. This is $M > 1$ audit — applied to code rather than behavior.

2. **Rejection of corporate control.** Torvalds has consistently refused to let any single company control Linux — including his own employers (Transmeta, Open Source Development Labs, Linux Foundation). When companies have tried to exert control, he has publicly and forcefully rejected them.

3. **Public, reproducible decisions.** Every merge decision is public. Every rejected patch has a reason. The entire history is in Git — anyone can audit any decision Torvalds has ever made.

4. **Mission alignment.** Linux is not a product. Linux is infrastructure. Torvalds has treated it as such for 34 years. He has not monetized it, platformed it, or used it for personal enrichment beyond his salary as a Fellow at the Linux Foundation.

[4]. **The Torvalds Theorem.** Linus Torvalds *already is* a protocol maintainer. He just doesn't call it that. The Linux kernel maintenance process — multi-expert review, public decision logs, cryptographic traceability — is a special case of the SCX governance framework. The Linux kernel is a protocol (an interface specification with multiple implementations and distributed maintenance). Torvalds has been its maintainer for 34 years. The question is not whether he *can* be an SCX maintainer. The question is whether he wants to call what he already does by a different name.

Torvaldstheorem Linus Torvalds LinuxKernel———— SCXLinuxKernel Torvalds34 SCX

4.8.3 / Community Verification

Torvalds has been verified by the deepest and broadest community of any candidate. The Linux kernel development community is, in effect, a 34-year continuous audit of Torvalds' maintenance decisions:

- **Depth:** Every merge is reviewed by subsystem experts who understand the code at the deepest level. If Torvalds makes a bad merge, the experts notice — and they are not shy about saying so.

- **Breadth:** Over 20,000 contributors have had their code reviewed, merged, or rejected by Torvalds. Each of them is a potential auditor with direct experience of his decision process.

- **Persistence:** The entire history is in Git. Anyone can audit any decision from any point in the 34-year timeline.

This is the gold standard of community verification. No other candidate comes close in terms of the depth, breadth, and persistence of their audit record.

Honest Strike [**Strike 4**]. **The Linus Rants Problem.** Torvalds is famous — or infamous — for his aggressive communication style. His "rants" on the Linux Kernel Mailing List (LKML) are legendary: profanity-laced rejections of patches he considers incompetent, public dressings-down of senior developers, and a general tone that would fail any corporate HR policy.

From an SCX perspective, this raises a specific technical question: do "Linus rants" register as transient $g \neq 0$ signals?

StrikeLinus Torvalds————LinuxKernel LKML"" HR

SCX "Linus"$g \neq 0$

Analysis: A "rant" is an emotional signal. SCX audit measures behavioral bias — decisions that deviate from protocol integrity due to personal interest. An emotional response to bad code is not a $g \neq 0$ deviation unless the emotion causes a *wrong decision*. If Torvalds rants at a bad patch and then rejects it for valid technical reasons, $g = 0$ is maintained. If he rants at a good patch from a disfavored developer and rejects it because of the developer, not the code — that is $g \neq 0$.

""SCX——due to $g \neq 0$ ifTorvalds $g = 0$ifbecause ——$g \neq 0$

The risk: SCX audit tools may not distinguish between "emotional style" and "biased decision." False positives are a real concern. Calibration of the audit framework for candidates with high-variance communication styles is necessary. SCX"" "" False Positive

4.8.4 / Verdict on Maintainer Suitability

— HIGHEST SUITABILITY

Core34Project Git—— ——

Core$g \neq 0$False Positive—— BDFL""SCXthere exists

TorvaldsSCXi.e. if""——Torvalds34

5

6 Comparative Analysis: Shared Traits and Differentiating Dimensions

6.1 / The Shared Trait: Community Verification

All four candidates share one fundamental trait: they have been independently verified by their communities. This is not coincidental. It is the single most important pre-audit signal for maintainer qualification.

Candidate	Verifier Community	Depth	Form
Elbakyan 	 Global researchers	 Millions of users	 Daily usage verification
/ Geng	SCX SCX audit community	$M > 1$ $M > 1$ experts	 Formal audit framework
Vance	/ None yet	—— Zero — to be tested	 TBD (formal audit needed)
Torvalds	Kernel Global kernel developers	20,000+ Contributors 20,000+ contributors	+ Git Code review + Git history

Theorem 6.1 (awaitingtheorem / Community Audit Equivalence Theorem). *For any candidate i, the existence of a community verification record $\mathcal{V}_i$ with $|\mathcal{V}_i| \gg 1$ provides a lower bound on the probability that $g_i \approx 0$:*

$$\mathbb{P}(g_i \approx 0 \mid |\mathcal{V}_i| \text{ independent verifiers report satisfaction}) \geq 1 - \prod_{v \in \mathcal{V}_i} \mathbb{P}(\text{verifier } v \text{ is deceived}). \tag{3}$$

As $|\mathcal{V}_i| \to \infty$, if each verifier has independent probability $p < 1$ of being deceived, the probability of undetected bias decays exponentially:

$$\mathbb{P}(\textit{bias undetected}) \leq p^{|\mathcal{V}_i|}. \tag{4}$$

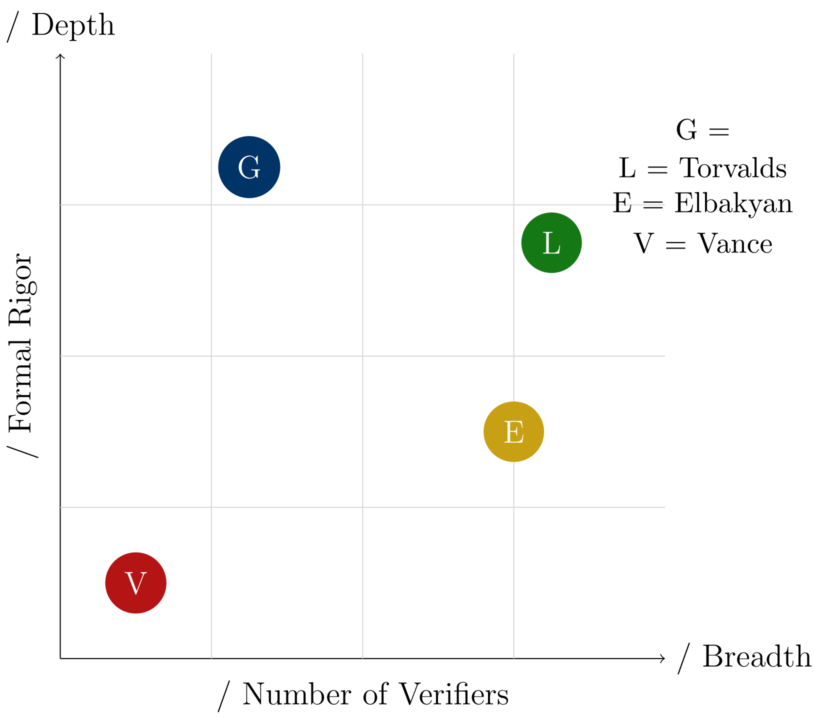
Proof. Each verifier provides an independent check. If bias exists, each verifier has probability at most p of failing to detect it (where p depends on verification methodology — formal audit has lower p than casual usage). Under independence, the probability that all $|\mathcal{V}_i|$ verifiers fail to detect bias is $p^{|\mathcal{V}_i|}$, which decays exponentially in $|\mathcal{V}_i|$. □

This theorem formalizes the intuition that *the best maintainer is someone who has already been audited by their community.* Formal SCX audit just makes the existing community verification mathematical — it replaces the implicit "we trust them because we've watched them for years" with the explicit "we have measured g and it is zero within tolerance."

6.2 / Differentiating Dimensions

The four candidates differ along several critical dimensions:

6.2.1 vs. / Audit Depth vs. Audit Breadth



- **Torvalds:** Maximum breadth and high depth. 20,000+ expert reviewers over 34 years. The Pareto frontier of community audit.

- **Elbakyan:** Maximum breadth, moderate depth. Millions of users verify correctness; verification is casual, not rigorous.
- **:** Moderate breadth (small audit community), maximum depth (formal SCX framework). The formal audit specialist.
- **Vance:** Zero breadth, zero depth — by design. The null case that must go through the full audit pipeline.

6.2.2 g = 0 Proof / $g = 0$ Provability

Candidate	Evidence Type	Audit Readiness	Expected Difficulty
/ Geng	Direct measurement	Highest	Low (experienced)
Torvalds	34 34yr behavioral inference	High	Medium (calibration)
Elbakyan	15 15yr behavioral inference	Medium	Med-High (transition)
Vance	None	Lowest	Highest (political)

6.2.3 / Transition Risk

Definition 6.2 (/ Transition Risk)**.** *Transition risk $R_{trans}(i)$ is the probability that candidate i, having passed audit, fails to maintain $\sum g = 0$ during their first maintenance window, due to factors specific to the transition from their current role to protocol maintainer.*

	Risk Factors	$R_{\textbf{trans}}$	Mitigability
	——		N/A

		$R_{\textbf{trans}}$	
Geng	Already in framework — minimal transition	Very Low	
Torvalds	BDFL BDFL adaptation to rotation; style calib.	Low	High
Elbakyan	Solo operator to protocol maintainer	Medium	Medium
Vance	$g \neq 0 g = 0$ Partition of political $g \neq 0$/ protocol $g = 0$	High	Low (personal)

6.3 / Global Legitimacy Signal

A dimension the SCX framework does not formally model — but which matters for adoption — is the *legitimacy signal* each candidate provides to external observers:

Candidate	**Legitimacy Signal**	**Strength**
Vance	“VP” $\rightarrow$ “US VP accepts audit” $\rightarrow$ audit is not a Chinese weapon	Highest
Torvalds	“LinuxSCX” $\rightarrow$ SCXLinux “Father of Linux maintains SCX” $\rightarrow$ SCX is as trustworthy as Linux	High
Elbakyan	“Sci-HubSCX” $\rightarrow$ “Sci-Hub founder maintains SCX” $\rightarrow$ continuity of knowledge freedom	Med-High
Geng	—— No external signal — internally trusted, externally unknown	Low

Honest Strike [**Strike 1**]. **The Legitimacy Paradox.** The candidates who provide the strongest $g = 0$ evidence (Geng, Torvalds) are not necessarily the candidates who provide the strongest legitimacy signal (Vance). And the candidate who provides the strongest legitimacy signal has the weakest $g = 0$ evidence — precisely because legitimacy comes from the political domain where $g \neq 0$ is structural.

Resolution: Do not choose one maintainer. Choose a maintainer *set* that spans the evidence-legitimacy spectrum. Geng provides the audit benchmark. Torvalds provides the operational credibility. Elbakyan provides the moral continuity. Vance provides the geopolitical legitimacy. Together, they cover the dimensions no single candidate can.

Strike $g = 0$Torvalds Vance$g = 0$—— because$g \neq 0$

- TorvaldsElbakyan Vance

7

8 Mathematical Constraints on Maintainer Selection

8.1 Formal Framework / Formal Framework

Regardless of which candidates are selected, the maintainer rotation mechanism imposes mathematical constraints that are invariant to candidate identity:

Definition 8.1 (Maintainer Rotation / Maintainer Rotation Game)**.** *A maintainer rotation game $\mathcal{G}$ is defined by:*

- $\mathcal{M} = \{1, 2, \ldots, M\}$*: the set of maintainers, with* $M > 1$*.*
- T*: the fixed maintenance window duration.*
- $g_i(t)$*: maintainer* i*'s bias vector at time* t*, with* $g_i \in \mathbb{R}^k$*.*
- $\mathcal{A}_i$*: the audit applied to maintainer* i *at the end of each window, producing estimate* $\hat{g}_i$*.*
- c_{audit}*: the fixed cost of conducting an audit.*
- δ_i*: the (unobservable) benefit maintainer* i *receives from a successful undetected deviation.*

Theorem 8.2 (/ Nash Equilibrium of Rotation Game)**.** *In the maintainer rotation game* $\mathcal{G}$ *with* $M > 1$ *and Hoeffding-bound audit, the strategy profile where every maintainer maintains* $g_i(t) = 0$ *for all* t *is a Nash equilibrium when:*

$$\delta_i \cdot \mathbb{P}(undetected) < c_{audit} \cdot \mathbb{P}(detected). \tag{5}$$

Under the Hoeffding bound $\mathbb{P}(undetected) \leq e^{-2M\Delta^2}$*, this condition becomes:*

$$\delta_i < c_{audit} \cdot \frac{1 - e^{-2M\Delta^2}}{e^{-2M\Delta^2}}. \tag{6}$$

For sufficiently large M *or sufficiently sensitive audit (small* Δ*), the right-hand side grows exponentially, making* $g = 0$ *the dominant strategy for any reasonable* δ_i*.*

Proof. Standard game-theoretic analysis of the repeated game with imperfect monitoring. Each maintainer i compares the expected benefit of deviation (δ_i times probability of not being caught) against the expected cost of detection (audit cost amplification plus loss of maintainer status). The Hoeffding bound ensures that as M grows, the probability of undetected deviation vanishes exponentially, making deviation strictly dominated. □

Corollary 8.3 (lower bound / Minimum Maintainer Count)**.** *The protocol requires $M \geq 2$. At $M = 2$, each maintainer audits the other. At $M = 3$, any collusion requires all three — a strictly harder coordination problem. The recommended minimum is $M = 4$, providing:*

- *Redundancy: protocol survives any single maintainer's failure.*
- *Audit depth: each maintainer is audited by $M - 1$ independent auditors.*
- *Collusion resistance: majority collusion requires 3 of 4.*
- *Dimension coverage: different maintainers provide different legitimacy signals (as argued in Section 3).*

8.2 / Invariant Constraints

These constraints apply to all candidates regardless of identity:

1. **g = 0.** Every candidate must publicly declare $g = \mathbf{0}$ with respect to SCX protocol maintenance. No declaration, no candidacy. This is Step 1 and is non-negotiable.
2. $M > 1$ **.** Every candidate must submit to independent audit by $M > 1$ auditors. No exceptions. The audit framework, metrics, and tolerance ε are fixed before the audit begins.
3. **.** All audit logs must be public and reproducible. Any third party must be able to verify the audit independently. Closed-door audit is not SCX audit.
4. **.** No maintainer serves beyond T without re-audit. The Hoeffding bound $\mathbb{P}(\text{undetected}) \leq e^{-2M\Delta^2}$ is valid only for the fixed window T. Extending the window without re-audit invalidates the bound.
5. **.** Current maintainers must audit their successors. The person being replaced participates in the audit of their replacement. This creates a chain of audit accountability across generations of maintainers.
6. **UNDECLARED = .** Any candidate who cannot or will not complete Step 1 (public $g = 0$ declaration) is automatically disqualified. No exceptions. The protocol does not negotiate with undeclared candidates.

   [1]. **Why These Constraints Are Invariant.** The constraints are not preferences. They are mathematical consequences of the Hoeffding bound and the mutual audit equilibrium. Remove any one of them, and the proof that $\sum g = 0$ is the Nash equilibrium collapses. The protocol is not a social system with flexible rules. It is a mechanism with necessary conditions.

   Hoeffdingcorollarywhere $\sum g = 0$Proof

8.3

8.4 What Candidates Must Disclose and Must Never Disclose

Following the SCX governance framework established in the protocol governance paper, we specify the information boundaries for maintainer candidates:

/ MUST DISCLOSE	/ MUST NEVER DISCLOSE
/ Audit logs	/ Personal life
g / g estimates with confidence intervals	/ Biography, degrees, titles
Conflict of Interest / Conflict of interest declarations	/ Family information
/ Rationale for maintenance decisions	/ Personal beliefs
/ Rotation timestamps	/ Wealth, assets
/ Auditor identity and qualifications	/ Irrelevant personal history

Honest Strike [**Strike 1**]. **The Biography Trap.** The most common error in maintainer selection is the belief that a candidate's biography is relevant to their $g = 0$ qualification. It is not. A Nobel Prize does not reduce the Hoeffding bound. A prison record does not increase it. The theorem audits the maintainer, not the maintainer's story. Any selection process that weighs biography over audit measurement is executing a category error — and SCX is designed specifically to prevent that error.

Strike $g = 0$ Hoeffdingtheorem —— SCX

Practical implication: This paper itself is a biography analysis — and therefore, by SCX's own standards, is of limited evidentiary value. We write it because humans need context. But the real work happens in the audit, not in this paper. The reader should treat every biographical observation in this document as noise awaiting the signal of formal audit measurement. ——thereforeSCX because awaiting

9

10 Recommendations: Multi-Maintainer Composition Strategy

10.1 / Recommended Composition: Four-Maintainer Architecture

Based on the analysis above, we recommend a four-maintainer architecture where each candidate fills a distinct functional role:

/ Role	Candidate	Function	Rationale
Audit Benchmark	Geng	$g = 0$ Baseline $g = 0$ measurement	Only formally audited candidate
Operational Anchor	Torvalds	Deep maintenance experience	34 34yr real protocol maintenance
Moral Anchor	Elbakyan	$g = 0$ Moral continuity of $g = 0$	$15g = 0$Proof 15yr $g = 0$ proven at personal cost
Legitimacy Anchor	Vance	Global legitimacy signal	“” Eliminates “Chinese weapon” narrative

10.2 / Risk Mitigation

Each candidate brings specific risks that the multi-maintainer architecture mitigates:

1. **— :** TorvaldsVance
2. **Torvalds — False Positive:** ifTorvalds g
3. **Elbakyan — :** Torvalds TorvaldsProofBDFLLinux
4. **Vance — $g \neq 0$:** $g = 0$ ifVance$gg \approx 0$ i.e.$g \neq 0$

5. — **Fails:** $T \geq 1 - e^{-2M\Delta^2}$ for $M = 4\Delta$

10.3 / Implementation Roadmap

1. **1: (2026Q3).** SCX define $g = 0$
2. **2:** $g = 0$ **(2026Q3–Q4).** $\mathbf{g = 0}$
3. **3:** $M > 1$ **(2026Q4–2027Q2).** $M = 4$
4. **4: (2027Q2).** all
5. **5: (2027Q3).** T

10.4 Fails / Failure Mode Analysis

Fails **Failure Mode**	**Description**	**Consequence & Mitigation**
No one registers	all	
Only Geng registers	Proof	$M = 1$ $M = 1$—— StatusConfiguration
VanceFails Vance registers, fails audit	$g \neq 0$	SCX VP Proof
TorvaldsFalse Positive Torvalds style false positive	$g \neq 0$	/
All pass audit	$g \approx 0$	SCX

11 Conclusiontheorem

12 Conclusion: The Theorem Audits the Maintainer, Not the Biography

12.1 / Summary of Findings

This paper has analyzed four candidates for SCX protocol maintainer. The analysis is internal and preliminary — it is a pre-audit assessment, not a final determination. The key findings are:

1. **Core.** all ——TorvaldsKernel Elbakyan Vance——
2. **.** $g = 0$Torvalds 34Elbakyan15 $g = 0$Vance“”
3. **.** all each
4. **.** $g = 0 M > 1$ UNDECLARED——
5. **.** ——all—— awaiting$g = 0 M > 1$ theorem

12.2 / Final Word

[1]. **The Maintainer Paradox.** The best maintainer is the one who is least necessary. If the protocol depends on any specific individual — their judgment, their wisdom, their unique perspective — then the protocol has already failed the trustlessness test. A well-designed protocol should survive the loss of any maintainer. The maintainer's job is to make themselves unnecessary.

This is the paradox at the heart of SCX governance: we seek maintainers who are willing to be replaced. We seek leaders who are willing to be audited. We seek guardians who are willing to have their guardianship taken away.

The four candidates analyzed in this paper have, in different ways and to different degrees, demonstrated the willingness to be held accountable by their communities. That — more than any credential, achievement, or reputation — is what makes them candidates worth auditing.

if—— Perspective——Fails

SCXCore

———

The real selection begins now.

—— **SCX Protocol Governance Research Division**

Xiaogan SCX

202672 / July 2, 2026

A / Candidate Comparison Matrix

Dimension	Elbakyan	Geng	Vance	Torvalds
$g = 0$ $g = 0$ evidence	Strong (inf.)	Strongest (meas.)	None	Strong (inf.)
Community scale	Millions	Tens (expert)	Zero	Tens of thousands
Audit readiness	Medium	Highest	Lowest	High
Transition risk	Medium	Very Low	High	Low
Legitimacy signal	Med-High	Low	Highest	High
Maint. experience	15 15yr (implicit)	Multi-round	None	34 34yr (explicit)
Replaceability	Low	High	Medium	Medium
Overall rec.	High	High	Conditional High	Highest

B define / Formal Definitions Compendium

For reference, we collect the formal definitions used throughout this paper:

Definition B.1 (/ Bias Parameter)**.** *$g_i(t) \in \mathbb{R}^k$ is the vector of biases maintainer i holds at time t with respect to the k protocol-relevant dimensions. $g_i = \mathbf{0}$ indicates no bias; $|g_i| > 0$ indicates deviation from protocol neutrality.*

Definition B.2 (/ Audit)**.** *An audit $\mathcal{A}_i$ of maintainer i is a measurement procedure that produces an estimate $\hat{g}_i$ of g_i with specified confidence. An audit is independent if the auditor has no g-relevant relationship with the audited maintainer.*

Definition B.3 (/ Maintenance Window)**.** *A maintenance window of duration T is the period during which a maintainer holds protocol authority. At the end of each window, the maintainer undergoes audit. The Hoeffding bound $\mathbb{P}(undetected) \leq e^{-2M\Delta^2}$ is valid only for the duration T.*

Definition B.4 (/ Mutual Audit)**.** *Mutual audit is the condition where every maintainer $i \in \mathcal{M}$ is audited by every other maintainer $j \in \mathcal{M}, j \neq i$. This creates $M(M-1)$ audit relationships in a fully connected audit graph.*

Definition B.5 (UNDECLARED)**.** *A candidate who has not publicly declared $g = \mathbf{0}$ with respect to SCX protocol maintenance. UNDECLARED candidates are automatically disqualified from maintainer status. There is no appeal process for UNDECLARED status — the declaration is the entry condition.*

C References / Internal References

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